REFV-KG Explorer: Causal and Stakeholder-Oriented Exploration of Renewable Energy Policy Discussions

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Abstract

Policy discussions about renewable-energy projects involve interrelated causes, effects, and countermeasures as well as differing stakeholder perspectives. Examining these elements across many documents is difficult when causal statements, opinions, and their supporting passages must be inspected separately. We present REFV-KG Explorer, an interactive demonstration for exploring causal structures and comparing stakeholder opinions together with their source evidence. The interface reads JSON data retrieved in advance from REFV-KG using SPARQL and provides coordinated filtering, tabular inspection, and network visualization. It enables users to examine keyword-centered causal structures together with differing stakeholder opinions. The contribution is the exploration and comparison workflow; construction and evaluation of the underlying knowledge graph are outside its scope.

Keywords

knowledge graph, causal structure, stakeholder opinion, evidence exploration, renewable-energy policy

1. Introduction

Renewable-energy projects are discussed by residents, governments, developers, researchers, and other stakeholders whose concerns and priorities may differ. Relevant statements are distributed across council minutes, committee records, governmental publications, news articles, and stakeholder documents. Understanding a policy issue therefore requires more than locating documents: an analyst must trace claims about causes, effects, and countermeasures, identify who expressed related opinions, and inspect the passages on which those records are based.

REFV-KG is a knowledge graph constructed from heterogeneous Japanese renewable-energy policy documents; its construction and evaluation are reported in our previous work [1]. Its statement-level annotations were developed using RDF-star [2]. RDF 1.2 now defines *triple terms* for referring to triples in statement-level descriptions [3]. The data and construction resources are public, and the repository provides explicit licensing information.[1] This demo instead focuses on interactive use of the graph for (1) exploring causal structures and (2) comparing stakeholder opinions with supporting text.

The interface combines faceted filtering, tabular evidence inspection, and a causal/stakeholder network. Users can also switch among keyword granularities; the motivation is discussed in Section 3.2.

Related systems address complementary aspects: an SDG knowledge organization system supports multilingual semantic search and content linking [4]; Stakeholder Suite analyzes actors, topics, arguments, and stances in public debates [5]; and Linked Data visualization systems expose RDF resources and their relations [6, 7]. REFV-KG Explorer instead coordinates exploration of causal relations, stakeholder opinions, and supporting evidence derived from the same knowledge graph.

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[1]https://github.com/oecu-kozaki-lab/REFV-KG

2. Demo Scenario and Design Requirements

2.1. Scenario

The system supports participants in examining causal structures and comparing stakeholder perspectives in renewable-energy policy discussions. For example, when a wind-power facility is planned, a user concerned about health or noise can search for "wind power–health" or "wind power–noise." The system lists relevant causal statements, related stakeholders and opinions, and supporting passages, and provides a graph overview. Users can inspect source-document information for records of interest. The system provides material for reflection and comparison rather than determining which opinion is correct or producing a consensus.

2.2. Design Requirements

Based on the scenario above, we define two primary requirements for supporting participants in exploring a policy issue and comparing stakeholder perspectives.

R1: Exploration of causal structures. Users should be able to explore cause, effect, and countermeasure relations concerning a selected policy issue, rather than inspecting isolated keywords or individual documents. The interface should also provide an overview of how multiple causal statements are interconnected.

R2: Comparison of stakeholder perspectives and evidence. For each causal statement, the interface should present the stakeholders who expressed related opinions, the content of those opinions, and the corresponding supporting passages. This enables users to compare different perspectives while retaining access to their documentary context and evidence.

Because the original REFV-KG classes are fine-grained, related statements can be fragmented across many small groups. We therefore introduced an additional class organization whose granularity is more suitable for the exploration and comparison tasks in R1 and R2, as described in Section 3.2.

3. System Design

3.1. REFV-KG Data Model and Mapping to the Explorer

REFV-KG was constructed from 523 heterogeneous Japanese renewable-energy policy documents and contains 22,922 causal triples; 79.6% are associated with at least one stakeholder opinion [1]. Its construction framework extracts causal statements and stakeholder opinions and represents them together with the source passages on which they are based. Figure 1 shows a simplified view of this data model and its mapping to the Explorer.

A causal statement is represented as a subject–predicate–object triple, where the predicate is one of three relation types: *cause*, *effect*, or *countermeasure*. Each causal triple is linked to its supporting source text through `prop:text-txt_contents` and to a related opinion through `prop:relatedOpinion`. The opinion is further linked to the stakeholder who expressed it through `prop:speaker` and to its content through `prop:content` [1]. This statement-level representation enables the causal claim, its supporting evidence, and the corresponding stakeholder perspective to be retrieved together. RDF 1.2 calls an RDF triple used as an RDF term within another triple a *triple term* [3].

REFV-KG Explorer directly reflects this structure in its exploration workflow. As illustrated in Fig. 1, the subject, relation, and object of a causal statement provide the principal causal-structure filters and are shown together in the result table. Stakeholder information supplies additional filtering conditions, while the corresponding stakeholder and opinion content are displayed alongside the causal statement. The supporting source text can also be inspected from the same result record. In this way, the interface maps the statement-level structure of REFV-KG to coordinated filtering and inspection

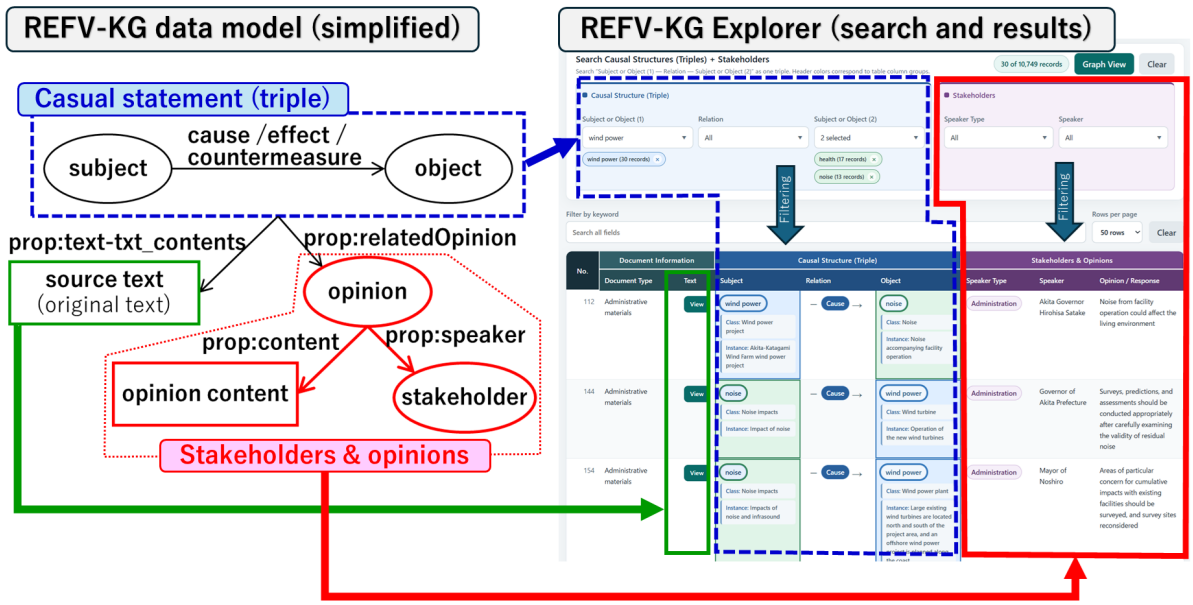


Figure 1: Mapping of REFV-KG causal statements (blue), supporting text (green), and stakeholder opinions (red) to Explorer filters and results.

of causal relations, stakeholder perspectives, and supporting evidence rather than exposing the graph only as isolated RDF resources.

3.2. Class Organization for Cross-document Exploration

REFV-KG contains 13,627 distinct keyword labels and 1,025 stakeholder labels, assigned to 7,832 keyword classes and 169 stakeholder classes. Because many original classes occur in only a few causal statements, they produced numerous small result groups and hindered cross-document overview.

We therefore introduced provisional intermediate classes. Keyword classes sharing the first two Japanese characters were grouped, the longest common prefix was used as the intermediate label, minor cases were manually adjusted, and groups occurring in only one or two statements were assigned to "Other." This reduced 7,832 keyword classes to 1,369 intermediate classes. Stakeholders were organized into five broad categories and ten intermediate types; OpenAI Codex with GPT-5.6 Sol assigned the 169 original classes to these predefined categories.

This redesign was motivated by formative feedback from project members with expertise in environmental policy and renewable energy who used an earlier version and found the original keyword classes too fine-grained. Because this was not a predefined user study and used no quantitative measures, we treat it as design input rather than formal evaluation. The intermediate classes are therefore provisional, and the original classes remain available.

3.3. Interface Overview and SPARQL-based Data Access

The interface is a browser-based application that reads JSON data retrieved in advance from REFV-KG using SPARQL; it does not query the endpoint directly from the user's browser. The retrieved data include causal statements, keyword classes, stakeholder classifications, related opinions, supporting text, and source-document information. This separation keeps the demo interface lightweight while preserving the graph structure needed for coordinated exploration. The extraction, generalization, and triple-term annotation processes used to construct REFV-KG are outside the scope of this demo. The REFV-KG data and construction resources are publicly available in a repository with explicit licensing information; the Explorer uses these data and augments them with the intermediate classes described in Section 3.2.

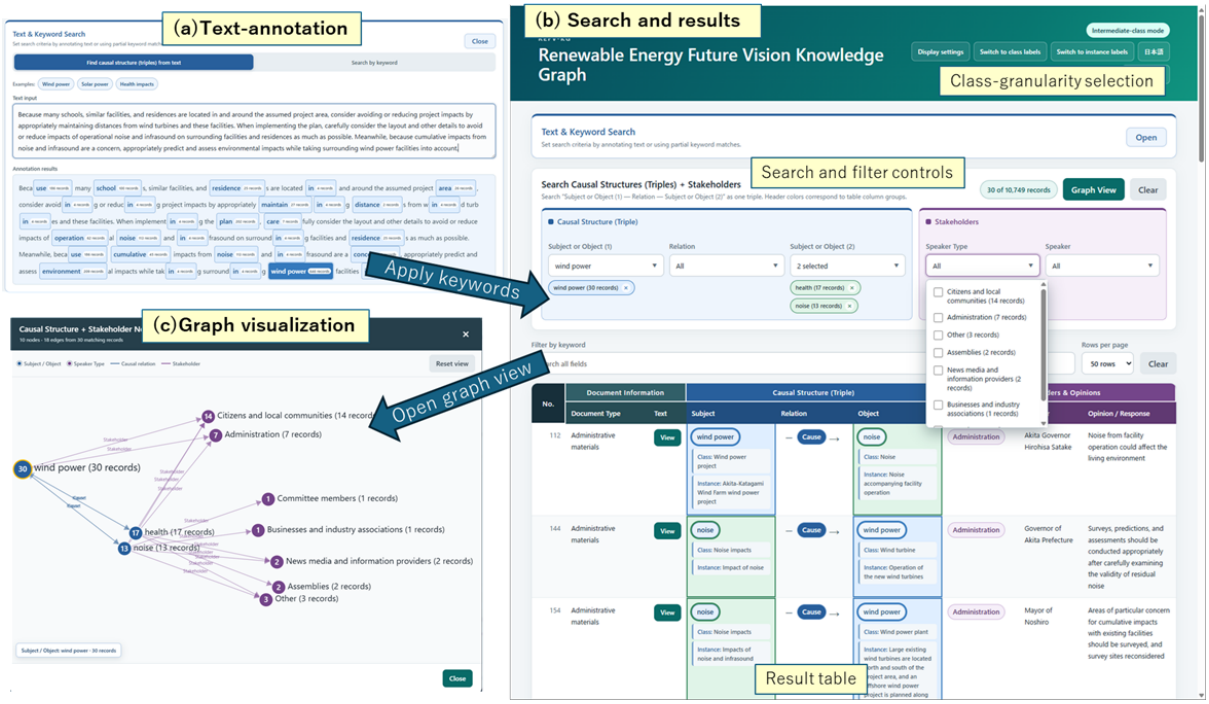


Figure 2: Overview and screen transitions of REFV-KG Explorer. Keyword annotations selected in the text-annotation screen (a) are applied to the search-and-results screen (b). Users can then open the graph view (c) to visualize the causal statements and stakeholders in the filtered results.

Users can specify keywords in two ways. They can search for keywords or keyword classes by partial string matching, or enter a discussion text and annotate it with keywords contained in the dataset (Fig. 2(a)). The latter function anticipates integration with an online discussion system: selected annotations are applied to the search screen, allowing users to retrieve causal statements and stakeholder opinions related to concepts in their own text.

The main screen combines keywords representing the causal relation of interest with stakeholder-related filters (Fig. 2(b)). By default, users can specify two keywords or keyword classes appearing as either the subject or object, a causal relation, a stakeholder type, and a speaker. Additional conditions, including prefecture, document type, and other stakeholder classifications, can be enabled in the display settings. Selected conditions are synchronously highlighted. The result table shows each statement as a subject–relation–object triple together with its supporting text, stakeholder, stakeholder type, and related opinion or response; additional document and classification fields can be displayed when needed.

At present, only the source filename is displayed because access to the original documents is subject to licensing restrictions, with external links planned for authorized project members. The original-class and intermediate-class views use the same underlying data, selected from the top menu; the intermediate-class view uses the added groupings for search while retaining the corresponding original-class information in the results.

3.4. Causal and Stakeholder Visualization

By clicking `Graph View`, users obtain a structural overview of all records in the filtered result set (Fig. 2(c)). The graph contains the keyword nodes selected as search conditions and the stakeholders who expressed related opinions, aggregated by their respective classes rather than displayed as individual instances for a compact overview.

Causal edges represent the three relation types defined in REFV-KG: cause, effect, and countermeasure. Stakeholder associations, derived from the triple-term annotations of individual causal statements in REFV-KG [1], are aggregated to connect the displayed causal structures with the corresponding stake-

holder classes; the count next to each node indicates the number of records matching the represented combination of causal relation and stakeholder condition.

The graph and result table support navigation between an overview and individual records: selecting a graph node filters the table to the relevant records, where users can inspect the causal statements, stakeholders, opinions, and supporting text. The `View` button displays information about the source document from which the corresponding triple was extracted. The graph aggregates all filtered results by class, although large result sets may reduce responsiveness.

3.5. Keyword Search at Different Class Granularities

Users can switch among intermediate classes, original REFV-KG classes, and individual keyword labels. The selected granularity changes the candidate values: for a partial match on "wind," the three modes yield seven, 53, and more than 200 candidates, respectively. Intermediate classes therefore provide a manageable default for cross-document exploration, while the other modes preserve finer-grained inspection. The corresponding original class is shown beside each intermediate-class label. Keyword granularity affects keyword search and filtering only; stakeholder classifications are selected independently.

4. Demonstration Walkthrough

The live demo uses *wind power*, *health*, and *noise* as search conditions, narrows results by stakeholder type, and inspects causal relations, opinions, and evidence in the table. We then open Graph View for an overview and inspect source-document information for selected records. We also demonstrate text annotation and allow participants to vary search terms, filters, and class granularity.

5. Design Rationale, Limitations, and Conclusion

The demo has three main limitations. First, the intermediate classes remain provisional and require systematic assessment for correctness, coverage, and task suitability. Second, transfer beyond the renewable-energy collection has not been validated. The Explorer itself operates on JSON data produced from SPARQL queries rather than depending on a fixed endpoint schema in the browser. It could therefore in principle be adapted to another knowledge graph that provides equivalent structures for causal relations, stakeholder opinions, and supporting evidence by replacing or adapting the SPARQL-to-JSON extraction layer. This is an architectural possibility rather than a demonstrated cross-domain result; validation with other knowledge graphs and domains remains future work. Third, no task-based user study has yet measured whether the interface improves issue understanding, comparison quality, or decision making.

The class organization also has limitations. Keyword grouping relies on short Japanese prefixes and minor manual adjustments, so lexically similar but semantically different classes may be merged, while synonymous classes with different prefixes may remain separate. The AI-assisted stakeholder assignments have likewise not yet been systematically validated.

REFV-KG Explorer contributes a workflow for jointly exploring causal structures, comparing stakeholder opinions, and checking supporting evidence in policy documents while keeping keyword granularity visible and adjustable. The contribution is the exploration workflow rather than the construction of REFV-KG itself, and the formative feedback described above is not treated as a formal evaluation. The online demo is available at https://refv-ex.kozaki-lab.jp/?lang=en, and its source code is publicly available under the Apache License 2.0 at https://github.com/oecu-kozaki-lab/REFV-KG-Explorer.

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Declaration on Generative AI

During the preparation of this work, the authors used OpenAI Codex to assist with implementing the application and assigning the original stakeholder classes to predefined intermediate and broad categories. The authors also used ChatGPT for translating author-written Japanese text into English, grammar and spelling checks, improving writing style, paraphrasing and rewording, and assisting with revisions of the manuscript in response to reviewer comments. All generated or revised content was reviewed and edited by the authors, who take full responsibility for the publication's content.

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